YANG CHEN PhD Candidate

yangchen@nnu.edu.cn | cubicsyang.github.io
School of Geography, Nanjing Normal University | Expected Jun. 2027
Wenzhou, Zhejiang, China | Born Oct. 1999

EDUCATION

Nanjing Normal University, School of Geography — Sep. 2021 – Present

PhD Candidate (integrated master's–doctoral program) in Cartography and Geographic Information Systems (GIS ranked No. 1 in China, ShanghaiRanking 2026); Advisor: Prof. Guo'an Tang

National University of Singapore, Urban Analytics Lab — Nov. 2024 – Nov. 2025

Visiting PhD Student funded by the China Scholarship Council (CSC); Advisor: Assoc. Prof. Filip Biljecki (Presidential Young Professor)

Zhejiang A&F University, College of Environment and Resources — Sep. 2017 – Jul. 2021

BSc in Geographic Information Science

RESEARCH INTERESTS

- **Vertical urban analytics:** Integrating multi-source geospatial data and artificial intelligence to characterize urban form, functional structure, and human–environment relationships, with emphasis on how vertical urban space affects human activity, urban extreme heat and microclimate, energy use, and sustainability.
- **Digital terrain analysis:** Combining remote sensing, GIS, and machine learning for large-scale terrain modeling, landform feature extraction, and landscape-evolution analysis, with emphasis on interactions among terrain, ecological processes, flood hazards, and human activity.

RESEARCH OUTPUTS

16 journal articles (5 first-author). Full list: Homepage, Google Scholar, ORCID. Metrics: 2025 JIF/JCR (released June 2026); Nature Index uses the 2026 journal list.

First-author Publications

1. **Y. Chen**, Z. Fan, H. Li, et al. Vision2Slope: Estimating urban road slopes with street view imagery. *International Journal of Geographical Information Science*, pages 1–45, 2026. DOI: 10.1080/13658816.2026.2656269 [JCR Q1; IF 5.9; Code]
2. **Y. Chen**, X. Yang, X. Li, et al. Vertical 15-minute city: Modeling urban density and functional mix with multi-source geospatial data. *Cities*, 169:106516, 2026. DOI: 10.1016/j.cities.2025.106516 [JCR Q1; IF 7.0; Nature Index]
3. **Y. Chen**, W. Sun, L. Yang, et al. Refining urban morphology: An explainable machine learning method for estimating footprint-level building height. *Sustainable Cities and Society*, 112:105635, 2024. DOI: 10.1016/j.scs.2024.105635 [JCR Q1; IF 13.3; Nature Index; Dataset; Code]
4. **Y. Chen**, X. Yang, H. Fu, et al. Characterizing spatial patterns and regionalization of anthropogenic landforms using multi-source geospatial data: Insights from the Loess Plateau of China. *Geomorphology*, 478:109708, 2025. DOI: 10.1016/j.geomorph.2025.109708 [JCR Q2; IF 3.2]
5. **Y. Chen**, X. Yang, L. Yang, et al. An automatic approach to extracting large-scale three-dimensional road networks using open-source data. *Remote Sensing*, 14(22):5746, 2022. DOI: 10.3390/rs14225746 [JCR Q1; IF 4.3; Code]

Selected Co-authored Publications

1. K. Zhou, **Y. Chen**, P. Liu, et al. A comprehensive vision-based framework for understanding nation-wide urban socio-demographics. *Cities*, 179:107657, 2026. DOI: 10.1016/j.cities.2026.107657 [JCR Q1; IF 7.0; Nature Index]
2. L. Yang, **Y. Chen**, Y. Li, et al. Is 3D building morphology really related to land surface temperature? Insights from a new homogeneous unit. *Building and Environment*, 266:112101, 2024. DOI: 10.1016/j.buildenv.2024.112101 [JCR Q1; IF 8.4]
3. P. Liu, **Y. Chen**, X. Liang, et al. A graph neural network for small-area estimation: Integrating spatial regularisation, heterogeneous spatial units, and Bayesian inference. *International Journal of Geographical Information Science*, 40(7):2358–2396, 2026. DOI: 10.1080/13658816.2025.2597971 [JCR Q1; IF 5.9; Code]

4. W. Sun, **Y. Chen**, X. Zhou, et al. Understanding the hydrological valley landscape: A multi-scenario adaptive framework for delineating valley floors. *Catena*, 256:109111, 2025. DOI: 10.1016/j.catena.2025.109111 [JCR Q1; IF 6.6; ArcGIS Pro Toolbox]
5. X. Zhou, **Y. Chen**, W. Sun, et al. Delineating individual alluvial fans and morphological analysis based on digital elevation models. *Geomorphology*, 473:109629, 2025. DOI: 10.1016/j.geomorph.2025.109629 [JCR Q2; IF 3.2]
6. H. Zhu, **Y. Chen**, X. Yang, et al. DEM-based sand dune analysis: Identification and morphological regionalization in the Grand Erg Oriental. *Earth Surface Processes and Landforms*, 50(1):e6042, 2025. DOI: 10.1002/esp.6042 [JCR Q2; IF 2.8]

Manuscripts in Preparation

1. **Y. Chen**, X. Yang, R. Zhu, et al. Urban topography amplifies the benefits of light-duty vehicle electrification.
2. M. Li[#], **Y. Chen**[#], X. Yang, and G. Tang. A global hierarchical typology of cities from multi-dimensional urban geomorphic contexts. ([#]Equal contribution)

Research Projects

PI, Jiangsu Postgraduate Innovation Program — Sep. 2023 – Sep. 2024

A Spatiotemporal Cube-Based Method for Geomorphic Process Analysis: led project execution and developed applications of a geographic spatiotemporal cube data structure for representing geomorphic processes.

Participant, National Natural Science Foundation of China (NSFC) Projects — Jan. 2020 – Present

- Key Program – *Theory and Methods of Digital Terrain Analysis Oriented Toward the Foundations of Geomorphology*: automated landform classification and mapping.
- General Program – *DEM-Based Geo-Info-Tupu of Anthropogenic Landforms in Small Watersheds of the Loess Plateau*: extraction and spatial heterogeneity of terraces, check dams, and settlements.

Participant, DDE Geomorphology Working Group — Jun. 2022 – Present

Designs storage and analysis workflows and a cloud visualization platform for geomorphological databases.

Academic Engagement & Service

- **Conferences**

4th International Conference on Urban Science and Intelligence – Guangzhou, China	Aug. 2026
2nd Academic Forum of Jiangsu Graduate Students (3MT talk) – Nanjing, China	Dec. 2025
AGU Fall Meeting – San Francisco, USA (first-author abstract; Dataset)	Dec. 2023
1st Conference on Information Geography – Suzhou, China	Nov. 2023
18th National Annual Conference on Geographic Information Science – Guilin, China	May 2023
City+ Conference – Perth, Australia	Sep. 2022

- **Peer review:** Reviewer for *Communications Earth & Environment*, *Computers, Environment and Urban Systems*, *International Journal of Geographical Information Science*, *Cities*, *International Journal of Digital Earth*, *npj Heritage Science*, *Scientific Data*, *Environment and Planning B: Urban Analytics and City Science*, and *Scientific Reports*.

Honors

National Scholarship, Ministry of Education of China	Sep. 2025
President's Scholarship, Nanjing Normal University (about 30 recipients university-wide)	Jun. 2026
Excellence Award, 3MT Competition, 2nd Academic Forum of Jiangsu Graduate Students	Dec. 2025
Excellent Ph.D. Research Plan, Nanjing Normal University	2025

Skills

- **Teaching:** Co-developed and taught the micro-credential course *Spatiotemporal Big Data Acquisition and Cloud Platforms* (54 class hours); teaching assistant for *Principles of GIS* and other courses.
- **Languages:** Chinese (native); English (CET-6, proficient in listening, speaking, reading, and writing).
- **Technical:** ArcGIS, QGIS; Python, C#, SQL; WebGIS, GeoAI.